

Abhishek Nilesch Shah

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Work Experience

Meta Platforms, Inc

NEW YORK, NEW YORK

May '21 – Present

Senior Software Engineer - Instagram and Threads Recommendations

- **Instagram Search** - Pioneered budget-efficient LLM based video understanding features for Instagram Search ranking and keyword suggestions to **improve search engagements in the US by 1%** and saving team's **GPU budget by 40%**.
- Improved relevance of search results by 2% by migrating relevance modeling stack from masked language model to large language model using video understanding features, auto-evaluations and agentic harness.
- **Threads** - Led a x-functional team to conceptualize, build and **launch Trending Now Surface for Threads** in [Japan](#) and worked on shipping the first version of [US](#) trends by working on LLM powered realtime and engaging trend detection pipeline driving Threads Search page DAU by 10% in both the countries.
- **Content Quality Foundations:** Led reels recommendation feature enhancement on user language matches and user interest matches driving content view in the user language to 100% improving reels engagements by working across XFN team of 10 engineers and 3 product managers. Reduced reels play counts to promote the content from original creators by reducing views on reposted content by 6%.
- Developed various ranking algorithms to enhance the followers of the creator accounts by working on user suggestion and foundational products of add yours in Instagram stories.

IBM Watson

NEW YORK, NEW YORK

Jun '17 – Apr '21

Advisory AI Software Engineer - Watson Assistant Core Algorithms

- Led development of assistant feature [show recommendations feature](#) enabling users to bootstrap conversational agents efficiently by searching chat logs. Implemented tools to support content-curation and management across 6 languages and 4 industrial domains hosted as part of [Content Catalog](#) in Watson Assistant.
 - Improved and maintained microservices for [Natural Language Understanding](#) supporting intent detection and entity extraction that work across 13 languages.
 - Designed and developed [universal language entity extraction](#) and [spellcheck service](#) for Watson Assistant to auto-correct incoming user sentences.
 - Developed a prototype of history store for service desk and channel-based [integrations](#).
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Additional Experience

IBM Watson

AUSTIN, TEXAS

Software Engineer Intern - Built telecom focused Conversational Agents for clients

Jun '16 – Aug '16

Hewlett Packard HP

BANGALORE, INDIA

R&D Engineer - Prototyped laserjet firmware migration from Windows to Ubuntu

Aug '14 – Jun '15

Information Retrieval Lab, DA-IICT

GANDHINAGAR, INDIA

Research Assistant - Worked on Temporal Information Retrieval and Indic Language Similarity

May '13– May '14

Checkdeck

GANDHINAGAR, INDIA

Software Engineer Intern - Built News Clustering System

May '12– Jun '12

Education

New York University (NYU), Courant Institute of Mathematical Sciences

NEW YORK, NEW YORK

M.S., Computer Science

Sep '15 – May '17

Dhirubhai Ambani Institute of Information and Communication Technology (DA-IICT)

GANDHINAGAR, INDIA

B.Tech., Information and Communication Technology

Aug '10– May '14

Noted Publications

- Lin Pan, Chung-wei Hang, Haode Qi, **Abhishek Shah et. al.**: Multilingual BERT Post Pretraining Alignment [\[pdf\]](#)
 - Tahira Naseem, **Abhishek Shah et. al.**: Rewarding Smatch: Transition-Based AMR Parsing with Reinforcement Learning, Paper accepted at ACL, Florence, 2019 [\[pdf\]](#)
 - Neil Mallinar, **Abhishek Shah et. al.**: Bootstrapping Conversational Agents With Weak Supervision, Emerging Application Track at AACL, 2019 (Co-presented at AACL, Hawaii, US, 2019) [\[pdf\]](#)
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Skills

Programming Languages: Python, SQL, HackLang, Java, C++, Javascript (using Typescript)

AI and ML: PyTorch, vLLM, SilverTorch, Claude, Llama, Agentic AI, Gemini, Reinforcement Learning

Data and Infrastructure: ElasticSearch, Apache Kafka, Apache Arrow, Hive, Spark, Node.js, Protobuf, Pandas, Sklearn, Kubernetes, Docker, PostgreSQL, Unicorn

Awards: Received IBM Patent Issuance Award and Plateau award. (Given for filing 4 or more patents)